Assignment

Statical and Hypothesis Testing

Part A

Q1. A factory has two machines, Machine A and Machine B. Machine A produces 60% of the total products, while Machine B produces 40%. The probability that a product produced by Machine A is defective is 5%, while the probability that a product from Machine B is defective is 8%.

- a) What is the probability that a randomly selected product is defective?
- b) If a randomly selected product is found to be defective, what is the probability that it was produced by Machine A?

Q2. There are three bags, Bag 1, Bag 2, and Bag 3. Each bag contains different colored marbles:

- Bag 1 has 3 red and 7 blue marbles.
- Bag 2 has 5 red and 5 blue marbles.
- Bag 3 has 6 red and 4 blue marbles.

One of the bags is chosen at random, and a marble is drawn at random from the selected bag. If the drawn marble is red, what is the probability that it was drawn from Bag 2?

Q3.A medical test is 95% accurate, meaning that:

- If a person has the disease, the test correctly identifies it 95% of the time.
- If a person does not have the disease, the test correctly identifies this 95% of the time.

In a certain population, 1% of people have the disease. If a randomly selected person tests positive, what is the probability that the person actually has the disease?

Q4. A call center receives an average of 5 calls per minute. The number of calls follows a Poisson distribution.

- a) What is the probability that the center receives exactly 8 calls in a particular minute?
- b) Given that the center receives at least 5 calls in a particular minute, what is the probability that it receives exactly 8 calls in that minute?

Part B

- **Descriptive Statistics: Measures of Central Tendency**
  - Calculate the **mean**, **median**, and **mode** of the following dataset:
    12, 15, 11, 13, 17, 18, 12, 15.
- **Descriptive Statistics: Mean**
  - What happens to the **mean** when a constant value is added to each data point in a dataset?
- **Descriptive Statistics: Measures of Variability**
  - Define **range** and **variance**. How do they differ?
- **Descriptive Statistics: Standard Deviation**
  - Calculate the **standard deviation** for the following dataset:
    4, 8, 6, 5, 3, 7, 9.
- **Descriptive Statistics: Skewness**
  - Explain what it means if a dataset has **positive skewness**.
- **Descriptive Statistics: Kurtosis**
  - What is **kurtosis**, and what does a high kurtosis value indicate about a dataset?
- **Descriptive Statistics: Frequency Distribution**
  - Create a **bar chart** representing the frequency distribution of the following categorical data:
    Red: 5, Blue: 7, Green: 3, Yellow: 8.

- **Descriptive Statistics: Histogram**
    - Given a dataset of continuous values, create a **histogram** with an appropriate bin size:
      15, 22, 30, 25, 40, 18, 35, 27.
- **Descriptive Statistics: Interquartile Range (IQR)**
    - Calculate the **interquartile range (IQR)** for the following data:
      3, 7, 8, 5, 12, 10, 15, 6.
- **Inferential Statistics: Central Limit Theorem**
    - Explain the **Central Limit Theorem** and why it is important in statistics.
- **Inferential Statistics: Hypothesis Testing**
    - Define the difference between **one-tailed** and **two-tailed** hypothesis tests.
- **Inferential Statistics: Type I and Type II Errors**
    - What is a **Type I error** in hypothesis testing? Give an example.
- **Inferential Statistics: Confidence Intervals**
    - If the mean of a sample is 50 with a standard deviation of 5, calculate the **95% confidence interval** for a sample size of 100.
- **Regression and Correlation Analysis: Correlation**
    - Define the **correlation coefficient**. What does a correlation coefficient of **0** signify?
- **Regression and Correlation Analysis: Scatter Plot**
    - Create a **scatter plot** for the following paired data:
      X: 2, 4, 6, 8, 10
      Y: 5, 10, 15, 20, 25.
- **Descriptive Statistics: Standard Deviation and Variance**
    - Calculate the **variance** and **standard deviation** for the following dataset:
      2, 4, 6, 8, 10, 12.
      Explain the relationship between variance and standard deviation.

- **Descriptive Statistics: Skewness and Kurtosis**
  - A dataset has a **skewness** of **-0.8** and a **kurtosis** of **3.5**. Explain the shape of the data distribution.
- **Inferential Statistics: Hypothesis Testing**
  - You are testing the hypothesis that a new drug increases recovery speed. You conduct a **two-tailed test** at the **0.05 significance level**. The p-value from your test is **0.03**. What is your conclusion, and what type of error might you be making?
- **Inferential Statistics: Confidence Intervals**
  - A researcher claims that the average IQ score in a population is 110. You take a sample of 50 people and calculate a mean IQ of 108 with a standard deviation of 15. Construct a **90% confidence interval** for the mean IQ of the population. Does this confidence interval support the researcher's claim?

Regression and Correlation Analysis: Linear Regression

- Using the following dataset, perform a **simple linear regression analysis** and find the regression equation:
  X: 1, 2, 3, 4, 5
  Y: 2, 4, 5, 4, 5.
  Interpret the slope and intercept of the regression line.